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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2014 Trial Examination

FORM VI

MATHEMATICS EXTENSION 1

Friday 8th August 2014

General Instructions

- Reading time — 5 minutes
- Writing time — 2 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 70 Marks

- All questions may be attempted.

Section I — 10 Marks

- Questions 1–10 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II — 60 Marks

- Questions 11–14 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Eleven.
- Write your candidate number on this question paper and submit it with your answers.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Candidature — 120 boys

Examiner

DS

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

1

Which expression is equivalent to $\cos 2x$?

- (A) $\sin^2 x - \cos^2 x$
- (B) $2 \sin^2 x - 1$
- (C) $2 \sin^2 x + 1$
- (D) $2 \cos^2 x - 1$

QUESTION TWO

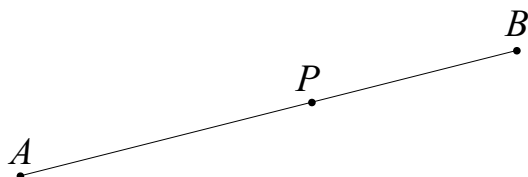
1

A polynomial of degree four is divided by a polynomial of degree two. What is the maximum possible degree of the remainder?

- (A) 3
- (B) 2
- (C) 1
- (D) 0

QUESTION THREE

1



In the diagram above the point P divides the interval AB in the ratio $3 : 2$. In what ratio does the point A divide the interval BP ?

- (A) $-5 : 3$
- (B) $-5 : 2$
- (C) $-3 : 5$
- (D) $-2 : 5$

QUESTION FOUR

1

What is the exact value of $\cos^{-1}(\cos(-\frac{\pi}{3}))$?

- (A) $-\frac{2\pi}{3}$
- (B) $-\frac{\pi}{3}$
- (C) $\frac{\pi}{3}$
- (D) $\frac{2\pi}{3}$

QUESTION FIVE

1

Which function is a primitive of $\frac{1}{1+4x^2}$?

- (A) $\frac{1}{2} \tan^{-1}(\frac{1}{2}x)$
- (B) $\frac{1}{4} \tan^{-1}(\frac{1}{2}x)$
- (C) $\frac{1}{2} \tan^{-1}(2x)$
- (D) $\frac{1}{4} \tan^{-1}(2x)$

Exam continues overleaf ...

QUESTION SIX

1

Which expression is equal to nC_2 ?

(A) $\frac{n}{2}$

(B) $\frac{n^2-n}{2}$

(C) $\frac{n^2+n}{2}$

(D) n

QUESTION SEVEN

1

The velocity v of a particle moving in a straight line is governed by the equation $v = x - 2$, where x is its displacement. The particle started at $x = 5$. What is the displacement function of the particle?

(A) $x = 5e^t$

(B) $x = 2 + \frac{1}{3}e^t$

(C) $x = 2 + e^t$

(D) $x = 2 + 3e^t$

QUESTION EIGHT

1

A particle is moving in simple harmonic motion about the origin according to the equation $x = 2 \cos nt$, where x metres is its displacement after t seconds. It passes through the origin with speed $\sqrt{2}$ m/s. What is the value of n ?

(A) $\frac{1}{\sqrt{2}}$

(B) $\sqrt{2}$

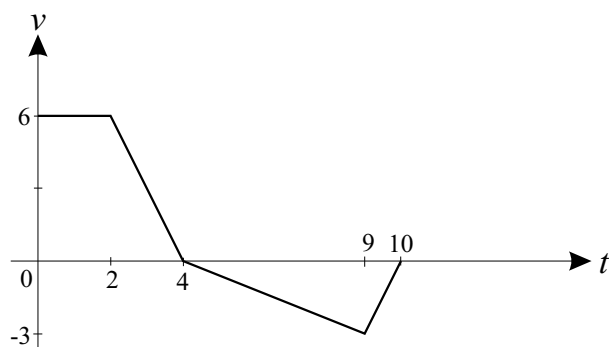
(C) $-\sqrt{2}$

(D) $\frac{\pi}{4}$

Exam continues next page ...

QUESTION NINE

1



The diagram above shows the velocity–time graph of an object that moves over a 10 second time interval. For what percentage of the time is the speed of the object decreasing?

- (A) 30%
- (B) 60%
- (C) 70%
- (D) It cannot be determined from the graph.

QUESTION TEN

1

How many solutions does the equation $2x + 3\pi \sin x = 0$ have in the domain $0 \leq x \leq 2\pi$?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

_____ End of Section I _____

Exam continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet. **Marks**

(a) Solve the inequation $\frac{2}{x} < 3$. 2

(b) (i) Sketch the curve $y = \sin^{-1} x$. 1

(ii) What is the gradient of the curve at $x = 0$? 1

(c) Solve the equation $\sin 2x = \sin x$ for $-\pi \leq x \leq \pi$. 3

(d) A curve is defined parametrically by the equations 2

$$x = 1 - t$$

$$y = t^2.$$

Find the gradient of the tangent to the curve at the point where $t = -3$.

(e) By using the substitution $u = \sin x$, or otherwise, evaluate $\int_0^{\frac{\pi}{4}} \sin^5 x \cos x \, dx$. 3

(f) A spherical balloon, with volume given by the formula $V = \frac{4}{3}\pi r^3$, is being filled with air at the constant rate of $200 \text{ cm}^3/\text{s}$. At what rate is its radius r increasing at the instant when it is 7 cm ? Give your answer correct to three significant figures. 3

QUESTION TWELVE (15 marks) Use a separate writing booklet.

Marks

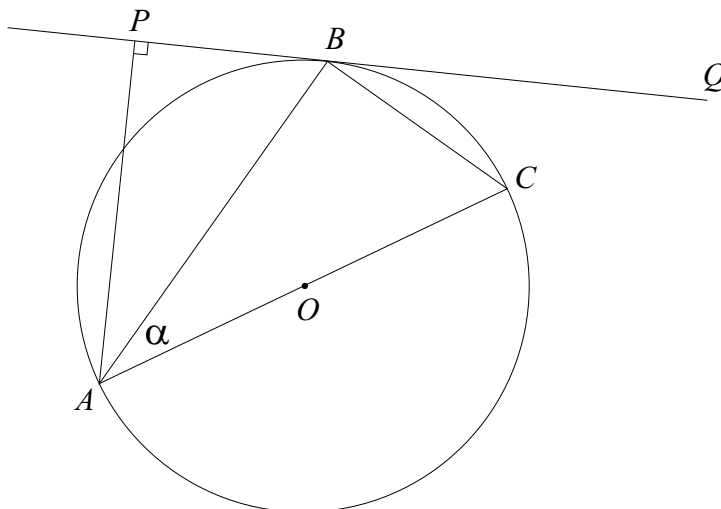
- (a) The cubic equation $x^3 + 5x^2 + cx + d = 0$ has three real roots -3 , 7 and α .
- (i) Use the sum of the roots to find α . 1
- (ii) Find the values of c and d . 2
- (b) Find the coefficient of x^3 in the expansion of $\left(3x^2 - \frac{2}{x}\right)^9$. 3
- (c) (i) Write the expression $\sqrt{2}\sin x - \sqrt{6}\cos x$ in the form $A\sin(x - \theta)$, where $A > 0$ and $0 < \theta < \frac{\pi}{2}$. 2
- (ii) Hence write down the maximum value of $\sqrt{2}\sin x - \sqrt{6}\cos x$, and find the smallest positive value of x for which this maximum value occurs. 2
- (d) Let $P(x) = x^3 + 3x - 7$.
- (i) Show that the equation $P(x) = 0$ has a root between 1 and 2. 1
- (ii) Use two applications of Newton's method with initial approximation $x_1 = 1$ to approximate this root. Give your answer correct to two decimal places. 2
- (e) Suppose that θ is the acute angle between the lines $y = kx$ and $(k + 1)y = kx$, where $k + 1 > 0$ and $k \neq 0$.
- (i) Find an expression for $\tan \theta$ in simplest form. 1
- (ii) Explain why $\theta < 45^\circ$. 1

QUESTION THIRTEEN (15 marks) Use a separate writing booklet.

Marks

(a)

3



The diagram above shows the points A , B and C lying on a circle, of which AC is a diameter. The line AP is perpendicular to the tangent at B .

Let $\angle BAC = \alpha$.

Prove that BA bisects $\angle PAC$.

- (b) A particle is moving in simple harmonic motion. Its acceleration is defined by the equation $\ddot{x} = -9x$. Whenever the particle is 4 cm from the origin its speed is 6 cm/s. Find the amplitude of the motion.

2

- (c) Consider the quadratic polynomial $Q(x) = (x + h)^2 + k$, for some constants h and k . Find the values of h and k given that $x + 2$ is a factor of $Q(x)$ and 16 is the remainder when $Q(x)$ is divided by x .

3

- (d) Prove by mathematical induction that for all positive integer values of n ,

3

$$1^2 \times 2 + 2^2 \times 3 + 3^2 \times 4 + \cdots + n^2(n + 1) = \frac{1}{12}n(n + 1)(n + 2)(3n + 1).$$

QUESTION THIRTEEN (Continued)

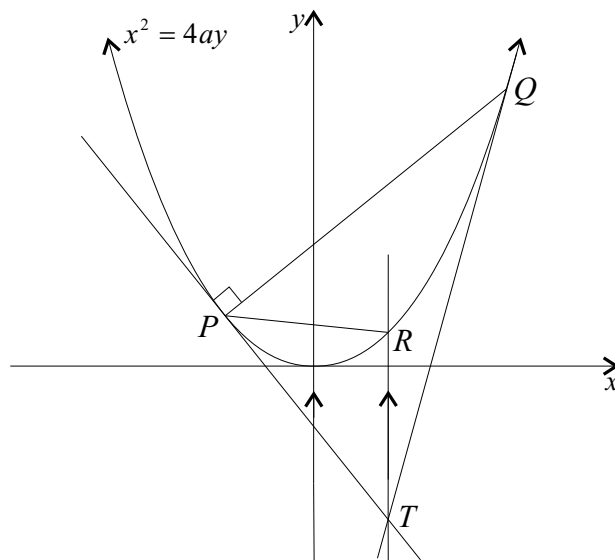
- (e) A jug of cold water at $W^\circ\text{C}$, where $W > 0$, is taken out of a refrigerator. The air temperature in the room is $2W^\circ\text{C}$. The rate at which the water warms is proportional to the difference between the temperature of the surrounding air and the temperature of the water. Thus $\frac{dT}{dt} = k(2W - T)$, where $T^\circ\text{C}$ is the temperature of the water after t minutes.

- (i) Show that $T = 2W - We^{-kt}$ satisfies the differential equation. 1
- (ii) If the temperature of the water has increased by 50% after 20 minutes, find the value of k . 2
- (iii) Find the percentage increase in the temperature of the water 45 minutes after the water is taken out of the refrigerator. Give your answer correct to the nearest whole percent. 1

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.

Marks

(a)



In the diagram above the normal at $P(2ap, ap^2)$ on the parabola $x^2 = 4ay$ meets the parabola again at $Q(2aq, aq^2)$. You may assume that the normal at P has equation $x + py = 2ap + ap^3$.

- (i) Show that $p^2 + pq + 2 = 0$. 2
- (ii) Given that the tangents at P and Q intersect at the point $T(a(p+q), apq)$, and the line through T parallel to the axis of the parabola meets the parabola at $R(2ar, ar^2)$, prove that PR is a focal chord. (That is, prove that $pr = -1$.) 2

Exam continues overleaf ...

QUESTION FOURTEEN (Continued)

- (b) (i) By considering the expansion of $(1+x)^n$, show that

1

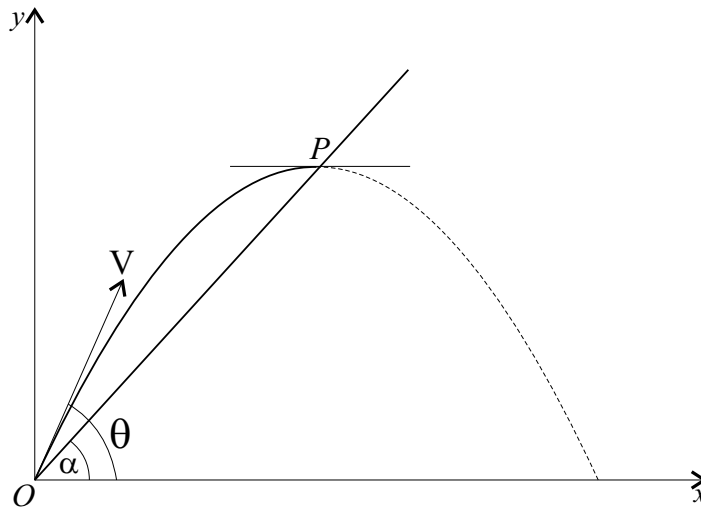
$$\binom{n}{1} + \binom{n}{2}x + \binom{n}{3}x^2 + \cdots + \binom{n}{n}x^{n-1} = \frac{(1+x)^n - 1}{x}.$$

- (ii) By applying integration to the identity in part (i), with the substitution $u = 1+x$ on the right-hand-side, show that

3

$$\binom{n}{1} - \frac{1}{2}\binom{n}{2} + \frac{1}{3}\binom{n}{3} - \cdots + \frac{(-1)^{n-1}}{n}\binom{n}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}.$$

- (c)



In the diagram above the point O is the foot of a plane inclined at an angle α to the horizontal. A particle is projected with speed V from O at an angle of elevation θ to the horizontal, where $\theta > \alpha$. It strikes the inclined plane at P , which is the vertex of the parabolic path of the particle. You may assume that this parabolic path has parametric equations $x = Vt \cos \theta$ and $y = Vt \sin \theta - \frac{1}{2}gt^2$.

- (i) Show that $\tan \theta = 2 \tan \alpha$.

3

- (ii) Show that the distance OP is given by $\frac{2V^2 \sec \alpha \tan \alpha}{g(1 + 4 \tan^2 \alpha)}$.

4

————— End of Section II —————

END OF EXAMINATION

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

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CANDIDATE NUMBER

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐

Multiple Choice (one mark each.)

(1) $\cos 2x = 2\cos^2 x - 1$ (D)

(2) degree one (C)

(3) $-5:3$ (A)

(4) $\cos^{-1}(\cos(-\frac{\pi}{3}))$
 $= \cos^{-1}(\cos \frac{\pi}{3})$
 $= \frac{\pi}{3}$ (C)

(5) $\int \frac{1}{1+4x^2} dx$
 $= \frac{1}{4} \int \frac{1}{\frac{1}{4} + x^2} dx$
 $= \frac{1}{4} \cdot 2 \tan^{-1} 2x + c$
 $= \frac{1}{2} \tan^{-1} 2x + c$ (C)

(6) ${}^nC_2 = \frac{n!}{2!(n-2)!}$
 $= \frac{n(n-1)}{2}$
 $= \frac{n^2 - n}{2}$ (B)

(7) When $t=0$, $x=5$,
 so only A or D are possible.
 In D, $v = \frac{dx}{dt} = 3e^t = x-2$.

So it's (D).

(8) $x = 2\cos nt$
 $v = -2n \sin nt$

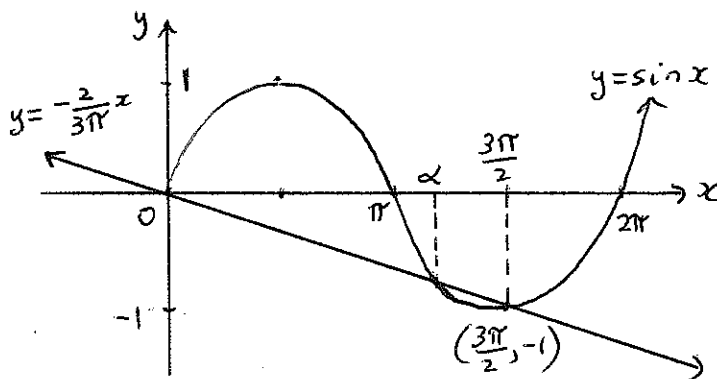
Maximum speed $= 2n = \sqrt{2}$.

So $n = \frac{1}{\sqrt{2}}$. (A)

(9) Speed decreasing
 for $2 < t < 4$ and $9 < t < 10$.

$\frac{3}{10} = 30\%$ (A)

(10) $2x + 3\pi \sin x = 0$
 $\sin x = -\frac{2}{3\pi} x$



Three solutions $x=0, \alpha, \frac{3\pi}{2}$.

(C)

Written Response

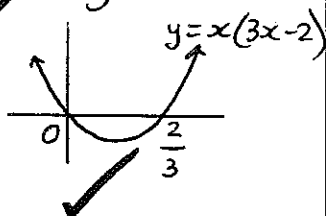
(11)(a) $\frac{2}{x} < 3, x \neq 0$

Multiply both sides by x^2 .

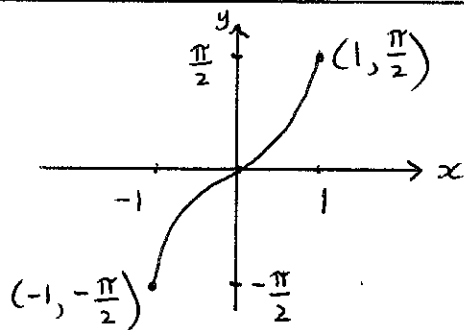
$2x < 3x^2$ ✓

$x(3x-2) > 0$

$x < 0$ or $x > \frac{2}{3}$ ✓



(b)(i)



(ii) At $(0,0)$ $m=1$. ✓

(c) $\sin 2x = \sin x, -\pi \leq x \leq \pi$

$2\sin x \cos x = \sin x$ ✓

$\sin x(2\cos x - 1) = 0$

$\sin x = 0$ or $\cos x = \frac{1}{2}$ ✓

$x = -\pi, 0, \pi, -\frac{\pi}{3}, \frac{\pi}{3}$

$= 0, \pm\pi, \pm\frac{\pi}{3}$ ✓

(d) $\begin{cases} x = 1-t \\ y = t^2 \end{cases}$

$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$

$= \frac{2t}{-1}$ ✓

At $t = -3, \frac{dy}{dx} = 6$. ✓

Alternatively eliminate t to get the Cartesian equation $y = (1-x)^2$.

Then $\frac{dy}{dx} = -2(1-x)$.

When $t = -3, x = 4$.

So $\frac{dy}{dx} = -2(-3) = 6$, as above.

(e) $u = \sin x$

$\frac{du}{dx} = \cos x$

$du = \cos x dx$

x	0	$\frac{\pi}{4}$
u	0	$\frac{1}{\sqrt{2}}$

$\int_0^{\frac{\pi}{4}} \sin^5 x \cos x dx = \int_0^{\frac{1}{\sqrt{2}}} u^5 du$
 $= \frac{1}{6} [u^6]_0^{\frac{1}{\sqrt{2}}}$
 $= \frac{1}{6} (2^{-\frac{1}{2}})^6$
 $= \frac{1}{6} \cdot 2^{-3}$
 $= \frac{1}{48}$ ✓

(f) Given $\frac{dV}{dt} = 200 \text{ cm}^3/\text{s}$.

$V = \frac{4}{3}\pi r^3$ so $\frac{dV}{dr} = 4\pi r^2$.

Find $\frac{dr}{dt}$ when $r = 7 \text{ cm}$.

$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$

$200 = 4\pi r^2 \cdot \frac{dr}{dt}$

$\frac{dr}{dt} = \frac{50}{\pi r^2}$

When $r = 7$,

$\frac{dr}{dt} = \frac{50}{49\pi} \text{ cm/s}$

$\approx 0.325 \text{ cm/s}$ ✓

(No penalty for incorrect rounding or incorrect number of figures.)

$$(12)(a)(i) -3 + 7 + d = -5$$

$$d = -9$$

$$(ii) \text{Product of roots} = -d,$$

$$\text{so } -3 \times 7 \times -9 = -d,$$

$$\text{so } d = -189.$$

$$\text{Sum of products of pairs} = c,$$

$$\text{so } -21 + 27 - 63 = c$$

$$\text{so } c = -57.$$

$$(b) \text{General term} = {}^9C_r \cdot (3x^2)^{9-r} \cdot \left(\frac{-2}{x}\right)^r$$

$$= {}^9C_r \cdot 3^{9-r} \cdot (-2)^r \cdot x^{18-3r}$$

$$\text{Let } 18 - 3r = 3$$

$$\text{Then } 3r = 15$$

$$r = 5.$$

$$\text{So the coefficient of } x^3 \text{ is}$$

$${}^9C_5 \cdot 3^4 \cdot (-2)^5 = -326592.$$

$$(c)(i) A \sin(x - \theta)$$

$$= A \sin x \cos \theta - A \cos x \sin \theta$$

$$= (A \cos \theta) \sin x - (A \sin \theta) \cos x$$

Equating the coefficients of $\sin x$ and $\cos x$, we get

$$A \cos \theta = \sqrt{2} \quad (1)$$

$$A \sin \theta = \sqrt{6} \quad (2)$$

Squaring and adding,

$$A^2(\cos^2 \theta + \sin^2 \theta) = 2 + 6$$

$$A = 2\sqrt{2} \quad (A > 0)$$

Dividing (2) by (1),

$$\tan \theta = \sqrt{3}$$

$$\theta = \frac{\pi}{3} \quad (0 < \theta < \frac{\pi}{2})$$

$$\text{So } \sqrt{2} \sin x - \sqrt{6} \cos x = 2\sqrt{2} \sin(x - \frac{\pi}{3})$$

(ii) So the maximum value is $2\sqrt{2}$, since $-1 \leq \sin(x - \frac{\pi}{3}) \leq 1$.

(continued) (ii) The maximum value first occurs when

$$x - \frac{\pi}{3} = \frac{\pi}{2}$$

$$x = \frac{5\pi}{6}$$

(d)(i) $P(1) = -3$ and $P(2) = 7$, so $P(x) = 0$ has a root between 1 and 2.

$$(ii) P'(x) = 3x^2 + 3$$

$$x_2 = x_1 - \frac{P(x_1)}{P'(x_1)}$$

$$= 1 - \frac{P(1)}{P'(1)}$$

$$= 1 - \frac{-3}{6}$$

$$= 1.5$$

$$x_3 = 1.5 - \frac{P(1.5)}{P'(1.5)}$$

$$= 1.5 - \frac{1.5^3 + 3(1.5) - 7}{3(1.5)^2 + 3}$$

$$= 1.410256 \dots$$

$$= 1.41 \text{ to 2 decimal places.}$$

$$(e)(i) m_1 = k \text{ and } m_2 = \frac{k}{k+1}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \quad (\theta \text{ is acute})$$

$$= \left| \frac{k - \frac{k}{k+1}}{1 + \frac{k^2}{k+1}} \right|$$

$$= \left| \frac{k^2 + k - k}{k+1 + k^2} \right|$$

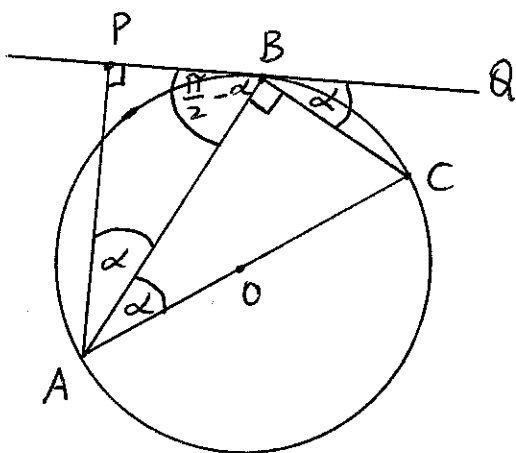
$$= \frac{k^2}{k^2 + k + 1} \quad (\text{since } k+1 > 0)$$

$$(ii) \frac{k^2}{k^2 + k + 1} < \frac{k^2}{k^2} \quad (\text{since } k+1 > 0)$$

$$\text{so } \tan \theta < 1 \quad (\theta \text{ is acute})$$

$$\text{so } \theta < 45^\circ$$

(13)(a)



✓ $\angle ABC = \frac{\pi}{2}$ (angle in semicircle)

✓ and $\angle CBQ = \alpha$ (alternate segment theorem)

✓ so $\angle ABP = \frac{\pi}{2} - \alpha$ (adjacent angles on a line)

so $\angle BAP = \alpha$ (angle sum of $\triangle BAP$)

so BA bisects $\angle PAC$.

(b) $\ddot{x} = \frac{d(\frac{1}{2}v^2)}{dx} = -9x$

so $\frac{1}{2}v^2 = \int -9x dx$
 $= -\frac{9}{2}x^2 + c$

When $|x|=4$, $|v|=6$.

So $18 = -72 + c$

$c = 90$.

So $\frac{1}{2}v^2 = 90 - \frac{9}{2}x^2$

$v^2 = 180 - 9x^2$ ✓

When $v=0$,

$x^2 = 20$

$x = \pm 2\sqrt{5}$ ✓

So the amplitude is $2\sqrt{5}$ cm.

OR use $v^2 = n^2(a^2 - x^2)$ ✓

(c) $x+2$ is a factor

so $Q(-2) = 0$

so $(-2+h)^2 + k = 0$

so $4 - 4h + h^2 + k = 0$ ① ✓

The remainder when dividing by x is 16

so $Q(0) = 16$

so $h^2 + k = 16$ ② ✓

② - ① : $4h - 4 = 16$

$h = 5$

Substitute into ② :

$25 + k = 16$

$k = -9$

So $h=5$ and $k=-9$. ✓

4

(13)(d) When $n=1$, $LHS = 1^2 \times 2$
 $= 2$

and $RHS = \frac{1}{12}(1)(2)(3)(4)$
 $= 2.$

So the result is true for $n=1$.

Suppose that the result is true for the positive integer $n=k$.
 i.e. suppose that $\sum_{n=1}^k n^2(n+1) = \frac{1}{12} k(k+1)(k+2)(3k+1)$. (*)

Prove that the result is true for $n=k+1$.

i.e. prove that $\sum_{n=1}^{k+1} n^2(n+1) = \frac{1}{12} (k+1)(k+2)(k+3)(3k+4)$.

$$LHS = \sum_{n=1}^k n^2(n+1) + (k+1)^2(k+2)$$

$$= \frac{1}{12} k(k+1)(k+2)(3k+1) + (k+1)^2(k+2) \quad (\text{using } *)$$

$$= \frac{1}{12} (k+1)(k+2) (k(3k+1) + 12(k+1))$$

$$= \frac{1}{12} (k+1)(k+2) (3k^2 + 13k + 12)$$

$$= \frac{1}{12} (k+1)(k+2)(k+3)(3k+4)$$

$$= RHS$$

So the result is true for $n=k+1$ if it is true for $n=k$.

But the result is true for $n=1$.

So, by induction, it is true for all positive integer values of n .

(e)(i) $T = 2W - We^{-kt}$

$$\left\{ \begin{aligned} \frac{dT}{dt} &= kWe^{-kt} \\ &= k(2W - T), \text{ as required.} \end{aligned} \right.$$

(ii) When $t=20$, $T = 150\%$ of W
 $= 1.5W$.

So $1.5W = 2W - We^{-20k}$

$$e^{-20k} = 0.5$$

$$e^{20k} = 2$$

$$20k = \ln 2$$

$$k = \frac{1}{20} \ln 2$$

(iii) When $t=45$,

$$T = 2W - We^{-45k}$$

$$= W(2 - e^{-45k})$$

$$= (1.789 \dots) W$$

So the temperature has increased by about 79%.

(14)(a)⁽ⁱ⁾ The coordinates $Q(2aq, aq^2)$ satisfy $x+py=2ap+ap^3$, (6)

$$\text{so } 2aq + apq^2 = 2ap + ap^3 \quad \checkmark$$

$$2q + pq^2 = 2p + p^3$$

$$2p - 2q + p^3 - pq^2 = 0$$

$$2(p-q) + p(p-q)(p+q) = 0$$

$$(p-q)(2 + p^2 + pq) = 0 \quad \checkmark$$

$p \neq q$, as P and Q are distinct,

$$\text{so } p^2 + pq + 2 = 0.$$

(ii) The vertical line TR has equation $x = a(p+q)$ and meets the parabola at $R(2ar, ar^2)$.

$$\text{So } a(p+q) = 2ar \quad \checkmark$$

$$p+q = 2r \quad \checkmark$$

Multiply both sides by p :

$$p^2 + pq = 2pr$$

But from (i), $p^2 + pq = -2$,

$$\text{so } -2 = 2pr$$

$$\text{so } pr = -1.$$

$$(b)(i) \quad (1+x)^n = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots + \binom{n}{n}x^n$$

Subtract 1 from both sides and then divide by x :

$$\frac{(1+x)^n - 1}{x} = \binom{n}{1} + \binom{n}{2}x + \binom{n}{3}x^2 + \dots + \binom{n}{n}x^{n-1} \quad \checkmark$$

$$(ii) \quad \int \frac{(1+x)^n - 1}{x} dx = \int \frac{u^n - 1}{u - 1} du \quad (\text{where } u = 1+x, \text{ so that } dx = du)$$
$$= \int (1 + u + u^2 + \dots + u^{n-1}) du$$
$$= u + \frac{1}{2}u^2 + \frac{1}{3}u^3 + \dots + \frac{1}{n}u^n + C, \quad \checkmark$$
$$= (1+x) + \frac{1}{2}(1+x)^2 + \frac{1}{3}(1+x)^3 + \dots + \frac{1}{n}(1+x)^n + C,$$

So integrating both sides of the identity in (a) gives

$$\binom{n}{1}x + \frac{1}{2}\binom{n}{2}x^2 + \frac{1}{3}\binom{n}{3}x^3 + \dots + \frac{1}{n}\binom{n}{n}x^n + C = (1+x) + \frac{1}{2}(1+x)^2 + \dots + \frac{1}{n}(1+x)^n \quad \checkmark$$

$$\text{Let } x=0, \text{ then } C = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}.$$

Let $x = -1$, then

$$-\binom{n}{1} + \frac{1}{2}\binom{n}{2} - \frac{1}{3}\binom{n}{3} + \dots + \frac{(-1)^n}{n}\binom{n}{n} + \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}\right) = 0$$

$$\text{so } \binom{n}{1} - \frac{1}{2}\binom{n}{2} + \frac{1}{3}\binom{n}{3} - \dots + \frac{(-1)^{n-1}}{n}\binom{n}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \quad \checkmark$$

(A better approach is to integrate both sides of (i) from -1 to 0 .)

(c)(i) $\dot{y} = V \sin \theta - g t$

At P, $\dot{y} = 0$, so $t = \frac{V \sin \theta}{g}$ ✓

So $x = \frac{V^2 \sin \theta \cos \theta}{g}$ and $y = \frac{V^2 \sin^2 \theta}{g} - \frac{1}{2} \cdot g \cdot \frac{V^2 \sin^2 \theta}{g^2}$
 $= \frac{V^2 \sin^2 \theta}{2g}$ ✓

Substitute these expressions into $y = x \tan \alpha$, the equation of OP:

$\frac{V^2 \sin^2 \theta}{2g} = \frac{V^2 \sin \theta \cos \theta}{g} \cdot \tan \alpha$
so $\frac{\sin^2 \theta}{2} = \cos \theta \tan \alpha$
so $\tan \theta = 2 \tan \alpha$ ✓

(ii) By Pythagoras, $OP^2 = (x\text{-value of P})^2 + (y\text{-value of P})^2$

$= \frac{V^4 \sin^2 \theta \cos^2 \theta}{g^2} + \frac{V^4 \sin^4 \theta}{4g^2}$
 $= \frac{V^4}{4g^2} (4 \sin^2 \theta \cos^2 \theta + \sin^4 \theta)$ ✓
 $= \frac{V^4}{4g^2 \sec^4 \theta} \left(\frac{4 \sin^2 \theta \cos^2 \theta}{\cos^4 \theta} + \frac{\sin^4 \theta}{\cos^4 \theta} \right)$ ✓
 $= \frac{V^4}{4g^2} \cdot \frac{4 \tan^2 \theta + \tan^4 \theta}{(1 + \tan^2 \theta)^2}$

BUT $\tan \theta = 2 \tan \alpha$,

so $OP^2 = \frac{V^4}{4g^2} \cdot \frac{16 \tan^2 \alpha + 16 \tan^4 \alpha}{(1 + 4 \tan^2 \alpha)^2}$ ✓

$= \frac{4V^4}{g^2} \cdot \frac{\tan^2 \alpha (1 + \tan^2 \alpha)}{(1 + 4 \tan^2 \alpha)^2}$

$= \frac{4V^4}{g^2} \cdot \frac{\sec^2 \alpha \tan^2 \alpha}{(1 + 4 \tan^2 \alpha)^2}$ ✓

So $OP = \frac{2V^2 \sec \alpha \tan \alpha}{g(1 + 4 \tan^2 \alpha)}$